

Simulation Is Not Adjudication

Institutional Admissibility and the Boundary Between Machine Output and Legal Consequence

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Abstract

AI systems increasingly produce outputs that emulate the form of legal interpretation, adjudicative reasoning, and compliance assessment. Existing governance frameworks address accuracy, explainability, and auditability, but leave under-articulated a prior question: whether a machine output can cross the boundary from technical event to institutional consequence. This paper argues, through conceptual analysis drawing on philosophy of mind (Seth, 2021; Searle, 1980), legal theory (Hart, 1961; Raz, 1979; Alchourrón and Bulygin, 1971), and AI governance (EU AI Act, Reg. 2024/1689), that simulation of legal decision-making is not institutional legal realization. It identifies four category errors that produce governance failures: treating optimization as interpretation, traceability as admissibility, local validity as aggregate legitimacy, and technical execution as institutional consequence. Drawing on the tradition of legal logic from Huerta Ochoa (2025), Bulygin (2015), Carrió (1965), Bobbio (1960), and Nino (1985), the paper argues that legal logic is not computational formalism: the distinction between norms, normative statements, and normative propositions shows that machine outputs, however logically coherent, occupy a different plane from the institutional act of norm application. The paper then proposes institutional runtime admissibility as a governance category positioned against existing doctrines of evidential admissibility, Hartian validity, and Razian authority, filling the gap none of them covers: the continuing condition that must remain valid at the moment a machine output is about to produce institutional consequence.

Keywords: Legal AI; algorithmic adjudication; runtime admissibility; institutional legitimacy; legal interpretation; AI governance; traceability; provenance; simulation; adjudication; EU AI Act; Hart; Raz; Alchourrón; Bulygin; legal logic; deontic logic; normative systems

1. Introduction

AI governance debates often focus on whether systems are accurate, explainable, auditable, safe, or aligned. These questions matter, but they leave a deeper legal problem under-articulated. As AI systems become capable of producing sophisticated outputs in legal, regulatory, administrative, and institutional contexts, they can emulate the external form of decision-making without performing the institutional act that gives a decision legal meaning.

The issue is not merely whether an AI system can generate a plausible legal interpretation, a coherent justification, a risk score, or an adjudicative recommendation. The issue is whether such an output can legitimately become a basis for reliance, action, or binding consequence within a legal or institutional order.

This paper argues that the distinction between simulation and realization is central to that problem, and that the tradition of legal logic offers a rigorous independent route to the same conclusion. A machine may simulate the linguistic and procedural form of legal reasoning. It may generate outputs that look like interpretations, decisions, authorizations, findings, or classifications. Yet the legal act of interpretation, adjudication, authorization, or commitment is not reducible to the production of a text-like or decision-like artifact. Legal meaning depends on authority, method, evidence, institutional recognition, contestability, and justified reliance within a normative order. And legal logic, as a discipline, has always known this: its entire history can be read as the sustained effort to distinguish what makes legal reasoning juridically valid from what makes it merely formally correct.

The core thesis is direct:

A simulation of legal decision-making is not itself institutional legal decision-making.

The governance question is therefore not only whether a model can produce an output that resembles a decision. It is whether the output can be admitted into a legal process as a legitimate basis for reliance or consequence.

2. Simulation Is Not Realization

Methodological note. This paper follows the same methodological move that characterizes the Law as Extended Phenotype research programme: importing a conceptual distinction from outside law, demonstrating that the structure of the distinction maps onto a legal problem, and deriving governance implications from that mapping. Seth's simulation/realization distinction was developed for philosophy of mind and consciousness studies. The translation to institutional legal theory is the author's own, and the argument for it appears below. The legal logic tradition, examined in section 2.1, provides an independent and internally generated route to the same conclusion.

Anil Seth's *Being You* (2021) offers a precise starting point. Arguing against functionalism in philosophy of mind, Seth distinguishes between simulating a process and realizing it. The example is elementary but exact: a computer simulation of weather, however detailed, does not get wet or windy. For some processes, simulation counts as instantiation. AlphaGo Zero is actually playing Go, not merely simulating it. But for processes whose relevant properties are not purely computational, simulation produces only the form of the process, not the process itself. Seth poses the question pointedly: is consciousness more like Go or more like the weather?

The argument has a direct implication for legal AI governance, though one Seth does not draw. Legal adjudication, interpretation, authorization, and institutional commitment are not purely computational processes. Their relevant properties are not exhausted by information processing. They depend on authority, method, source hierarchy, procedural posture, evidentiary constraints, institutional recognition, and justified reliance within a normative order. A machine output may emulate the linguistic and procedural form of a legal decision while lacking every one of these properties.

The translation from Seth's domain to the legal domain requires an independent step. It is not that machines lack consciousness and therefore cannot adjudicate. The point is structurally prior: legal acts are constituted by their relationship to a normative order, not by their surface form. A text that looks like a judgment is not a judgment. A classification that resembles a legal finding is not a legal finding. The institutional act of deciding requires, as H.L.A. Hart argued in *The Concept of Law* (1961), that a rule of recognition confer validity on the act, not merely that the act resemble a valid output.

Searle's Chinese Room (1980) is useful here, though for a different reason than Seth acknowledges. Seth excludes it from his main argument because it targets understanding rather than consciousness. For legal AI governance, that re-emphasis is precisely the point. The Chinese Room demonstrates that a system may produce syntactically correct outputs while lacking semantic grasp of what those outputs mean. Legal interpretation is not syntactic production: it requires understanding of normative hierarchy, institutional context, and consequential stakes. An AI system generating a legally plausible output is, in Searle's terms, manipulating symbols according to rules. The institutional act of interpretation requires more than that.

Epistemological positioning. This paper belongs to a research programme (Lakatos, 1978) that applies concepts from biology, philosophy of mind, and analytical legal theory to legal and institutional systems. Popper (1959) supplies the demarcation criterion: the governance failures identified here generate falsifiable predictions about institutional outcomes. The hard core of the programme is preserved while this paper extends the auxiliary belt to the domain of AI governance and legal admissibility.

2.1 Legal Logic Is Not Computational Formalism

The simulation/realization distinction has an independent and internally generated counterpart within the philosophy of law itself. The tradition of legal logic, far from supporting the assimilation of legal reasoning to formal computation, demonstrates precisely why such assimilation fails. Understanding this is important for AI governance: it closes off a reply that defenders of AI-driven legal systems might otherwise make, namely that legal logic is just a form of formal reasoning and that sufficiently sophisticated formal reasoning systems can therefore perform legal logic.

Legal logic should not be confused with computational formalism. As Huerta Ochoa (2025) shows in her systematic examination of the field, the term legal logic does not refer to a single operation but to several distinct relations between logic and law: formal reasoning applied to legal argument, deontic logic governing the structure of norms, logic of the legal system addressing coherence and consistency, and juridical argumentation theory. Each of these relations operates on different objects, uses different tools, and produces different types of output. A machine that preserves formal consistency may be performing well on one of these dimensions while entirely missing the others. The multiplicity is not terminological imprecision; it reflects a genuine complexity in the relationship between logical structure and juridical validity.

The decisive distinction is between norms, normative statements, and normative propositions. Huerta Ochoa, following Alchourrón and Bulygin, identifies three levels of legal language that must not be conflated:

1. Norms: prescriptive in character, accessible only through interpretation within a system, constituting the meaning of normative statements.
2. Normative statements: issued by a competent authority through a normative action, prescriptive in function, expressing the norm.
3. Normative propositions: describing the law, descriptive in function, susceptible to truth values.

A machine output, however sophisticated, occupies the third level. It describes, classifies, and infers about norms. It is not a normative statement, because it was not produced by a competent authority through a normative action. It is not a norm, because it does not constitute prescriptive meaning within a system. The distinction is categorical, not one of degree: formal correctness at level three does not generate validity at level two. No amount of computational sophistication changes the category.

The distinction between logical validity and legal validity is equally decisive. A machine-generated inference may be formally correct, traceable, and internally coherent, while still lacking legal validity if the norm it applies is not valid, current, applicable, authoritatively interpreted, or admissible within the relevant legal system. As Huerta Ochoa observes, following von Wright, legal validity is not a substitute for or parallel to logical truth: it is a relational property between a norm, the legal system, and a moment in time, determined by the rules of introduction that the

system itself provides. Legal consequence cannot therefore be derived from computational consistency alone.

This matters for AI governance in a specific way. It forecloses a move that governance frameworks sometimes implicitly make: treating formal correctness, traceability, or logical coherence as sufficient conditions for legal admissibility. They are not sufficient conditions. They are not even conditions of the same type. Formal correctness is a property of arguments. Legal validity is a property of norm membership in a dynamic system. Admissibility is an institutional judgment about whether an output may enter a legal process as a basis for consequence.

The dynamic character of the legal system adds a further dimension. Bulygin (2015) defines systemic validity as a triadic relation among a norm, a legal system, and a moment in time. A norm belongs to a system at a given moment in virtue of the system's rules of introduction at that moment. Authority may expire. Interpretive standards may shift. Applicable norms may be superseded. A machine output generated at one moment may no longer correspond to the valid, applicable norm at the moment of reliance or commitment. This temporal fragility is not a technical problem to be solved by better logging or more frequent model updates. It is a structural feature of legal normativity that requires ongoing institutional evaluation, not static clearance.

Carrió (1965), from within the rioplatense analytical tradition, makes the same point through the lens of legal language. There is a categorical difference between the prescriptive use of legal language, through which a judge or legislator creates law, and the descriptive use through which an analyst explains it. A system that generates a legally plausible interpretation is operating in the second mode. To operate in the first, it would require the institutional position that enables prescriptive use, which is precisely the condition no computational process can confer upon itself. In the zone of open texture that characterizes legal concepts, the interpretive decision is not the retrieval of a pre-determined answer but the performance of a constitutive act by a positioned institutional interpreter.

The point is not that logic is irrelevant to law. Legal logic has been, and remains, indispensable for the analysis of normative consistency, the detection of gaps and contradictions, and the assessment of argumentative validity. The point is that legal logic operates within a normative system whose validity conditions are not reducible to computation. What the tradition of legal logic reveals is not a blueprint for computational legal reasoning but a precise map of why the conditions for legal validity exceed the conditions for formal correctness.

3. Optimization Is Not Interpretation

A first category error arises when optimization is treated as interpretation.

AI systems can optimize for patterns, probabilities, rankings, classifications, semantic similarity, prediction, retrieval, or task completion. In legal contexts, such optimization may generate outputs

that appear interpretive. A system may summarize a statute, compare precedents, classify risk, identify inconsistencies, or recommend an action.

But legal interpretation is not merely a mapping from inputs to outputs. It occurs within a normative order. It presupposes source hierarchy, institutional method, accepted canons, procedural posture, evidentiary constraints, and an accountable interpreter. Neil MacCormick (1978) argued that legal reasoning is institutional reasoning: it takes place within systems of rules that determine not only what counts as a valid argument, but who is authorized to make it and in what procedural setting. The fact that an optimizer generates a legally plausible answer does not establish that the system has performed interpretation in the juridical sense.

Joseph Raz's account of authority (1979) sharpens the point. An authoritative legal directive gives subjects reasons for action that are not merely the same reasons they would have independently. Authority adds a normative layer: the directive counts as binding because it comes from a source recognized by the system as having the power to bind. An AI output lacks that normative layer entirely. It may be useful, even excellent, as analysis. It cannot by itself become authoritative.

This distinction is particularly relevant for regulatory frameworks that classify AI systems by intended purpose, risk, or function. Where an AI system generates outputs that appear to support legal, administrative, or adjudicative action, regulators may be tempted to treat the output as if it were an institutional interpretation. But a model operating as an optimizer does not possess legal intentionality or institutional authority merely because its output is decision-like.

The relevant governance question is therefore:

When may an optimized output be admitted as a legally relevant interpretation?

That question cannot be answered by performance metrics alone. It requires an admissibility inquiry.

4. Traceability Is Not Admissibility

A second category error arises when traceability is treated as a substitute for admissibility.

Traceability is indispensable. Legal and institutional systems must be able to reconstruct how a machine output was produced, what sources were used, which prompts or parameters were involved, which model version was active, what human interventions occurred, and what state transitions preceded the output.

But traceability does not itself confer legitimacy. It makes admissibility evaluable; it does not replace admissibility.

An output may be fully traceable and still inadmissible. The path may be reconstructible but normatively defective. The evidence may be logged but insufficient. The authority may be documented but expired. The procedure may be visible but improper. As Huerta Ochoa

emphasizes, the truth conditions applicable to propositions about a legal system are distinct from the validity conditions applicable to norms within that system: a system may produce coherent and traceable chains of normative propositions without any of those propositions constituting a valid norm or a valid normative statement. The governance principle is direct:

Traceability makes admissibility reviewable, but it does not make the output admissible.

This distinction matters directly for EU AI Act compliance. Article 13 of Regulation 2024/1689 requires sufficient transparency of operation. Article 12 mandates logging capabilities for post-market monitoring. These are traceability requirements. They are necessary conditions for accountability. Neither article addresses the normative question of when a traced output may serve as the basis for a legally binding consequence. The Act, at this point, treats traceability as if it were admissibility. That conflation is one of the structural gaps this paper identifies.

5. Local Validity Is Not Aggregate Legitimacy

A third category error arises when local validity is assumed to compose into aggregate legitimacy.

Complex AI and agentic workflows may involve many steps: retrieval, classification, summarization, comparison, escalation, delegation, scoring, approval, execution, logging, and handoff. Each step may appear locally valid. The system may have followed its rules. Permissions may have been checked. Evidence may have been recorded. State transitions may be consistent.

Yet a chain of locally valid steps can produce an aggregate consequence that was never evaluated as a whole.

This is especially important in machine-speed governance. A system may preserve procedural correctness at each micro-step while drifting away from the authority, context, evidence, mandate, or oversight conditions that originally justified the workflow. The resulting consequence may be traceable, locally valid, and technically executable, yet not legitimate within the surrounding legal or institutional order. The failure mode is precise:

Local validity must not launder aggregate consequence.

The chain itself may require admissibility review. It is not enough to ask whether each step was authorized. It is also necessary to ask whether the path from interpretation to reliance to commitment remained legitimate as a whole.

This problem has no parallel in the EU AI Act's current architecture. Article 14 requires that high-risk AI systems allow natural persons to oversee their functioning. But the oversight obligation is framed as a per-output or per-use requirement, not as a chain-level legitimacy review. A regulator checking each step for compliance may miss an aggregate consequence that no single step individually authorized.

6. Technical Event vs. Institutional Consequence

A fourth category error arises when technical execution is treated as institutional consequence.

A model output, API call, signed transaction, workflow transition, or automated recommendation may be a technical event. It becomes an institutional consequence only when it affects rights, obligations, status, access, liability, compliance position, evidentiary standing, or decision-making authority within a normative order. The boundary between technical event and institutional consequence is where governance becomes legally consequential.

At that boundary, the question is no longer merely whether the system could execute. The question is whether the consequence should be allowed to bind. This requires more than procedural validity. It requires authority, evidence, context, institutional recognition, and justified reliance at the point of commitment.

The distinction may be framed as a sequence of governance states:

1. Interpretation: a meaning-bearing output is generated or selected.
2. Reliance: a human, institution, or downstream system treats the output as a basis for action.
3. Commitment: the output or action produces a consequence that binds reality in a legal, institutional, economic, or operational sense.

The danger often lies in the transition between these states. A plausible interpretation does not by itself justify reliance. Justified reliance does not automatically authorize commitment. Commitment requires a separate admissibility condition.

7. Institutional Runtime Admissibility

7.1 Positioning Against Existing Doctrines

The concept of institutional runtime admissibility requires careful positioning against existing legal doctrines, each of which addresses an adjacent problem without covering the same ground.

Evidential admissibility, in common law systems, governs whether evidence may be presented before a tribunal (Federal Rules of Evidence, Rule 402; *Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 509 U.S. 579, 1993). In civil law systems, the analogous doctrines of *pertinencia* and *licitud* govern whether evidence meets relevance and procedural requirements. These doctrines apply after a legal process has already begun, to materials already produced. Institutional runtime admissibility operates earlier: it asks whether a machine output may enter a legal process at all, as a basis for reliance, before a tribunal or institution has been engaged.

Hartian validity (Hart, 1961) concerns whether a legal rule satisfies a system's rule of recognition and therefore counts as law. It is a criterion for the legal status of norms, not of outputs. A machine

output is not a norm; it is an artifact. Asking whether it is valid in Hart's sense is a category mistake. Institutional runtime admissibility asks a different question: not whether the output has the status of a legal rule, but whether it may support a legal consequence.

Razian authority (Raz, 1979) concerns whether a directive gives subjects reasons for action beyond those they would have independently. Authority is a property of sources, not of outputs. A machine output is not a source of authority in Raz's sense. Institutional runtime admissibility does not claim that machine outputs can be authoritative. It asks whether they may, under specifiable conditions, be admitted as legitimate inputs to institutional processes that remain authoritative in the human and institutional sense.

The tradition of analytical legal logic provides a fourth point of contrast, and in some ways the most illuminating one. Alchourrón and Bulygin's account of normative systems (1971/1993) defines a legal system as a set of statements whose deductive consequences include statements correlating cases with normative solutions. A machine output can reproduce the formal structure of such a system with high fidelity. What it cannot do is occupy the position of a normative statement within that system, because normative statements are introduced into the system through normative acts by competent authorities, not through computational generation. Bulygin's definition of systemic validity as a triadic relation among norm, system, and time (2015) makes clear that the question of whether an output is admissible as a basis for institutional consequence cannot be answered by inspecting the output alone: it requires evaluation of the system's current rules of introduction, the current authority conditions, and the current temporal validity of the applicable norms. That evaluation is exactly what institutional runtime admissibility provides.

Bobbio's distinction among validity, efficacy, and justice (1960) adds a complementary dimension. A machine output may be materially correct (just in Bobbio's sense) and practically effective (efficacious) while lacking validity: it was not produced by the competent organ through the correct procedure. Nino's distinction between descriptive and normative validity (1985) sharpens this further: even a descriptively valid output, one that accurately describes what a valid norm prescribes, lacks normative validity because normative validity requires justificatory reasons for compliance that do not emerge from the process of computational production.

Carrió (1965) supplies the linguistic register of the same argument: the prescriptive use of legal language through which law is created differs categorically from the descriptive use through which it is analyzed. No sophistication of descriptive performance converts description into prescription. The gap that runtime admissibility fills is therefore: the continuing condition that must be satisfied at the moment a machine output is about to produce institutional consequence, governing not its legal status as a norm, not its authority as a directive, not its evidential admissibility before a tribunal, and not its formal membership in a normative system, but its institutional legitimacy as a basis for reliance, action, or binding consequence within a normative order at a specific moment in time.

7.2 The Framework

Institutional runtime admissibility asks whether, at the moment a machine output is about to support reliance, action, or consequence, the relevant legal and institutional conditions remain valid. It is not a one-time clearance. It is a continuing condition that may decay over time or be interrupted by changes in authority, context, evidence, policy, oversight, mandate, or procedural posture.

This is especially important in systems that persist across interruption, restoration, delegation, handoff, reentry, or multi-agent composition. An output admissible when generated may no longer be admissible when execution resumes. A delegated authority may expire. A policy may change. Evidence may become stale. Oversight requirements may shift. The institutional context may no longer support reliance.

Institutional runtime admissibility therefore requires at least six inquiries:

1. Source authority: Are the legal or institutional sources used by the system valid and current? Failure example: a compliance system applies a regulatory standard superseded three months earlier; its outputs remain traceable but are inadmissible as a basis for compliance certification.
2. Methodological admissibility: Is the interpretive or analytical method acceptable in the relevant normative order? Failure example: a system trained on one jurisdiction generates risk classifications for a transaction governed by a different legal system; the method is not recognized in the relevant order.
3. Evidentiary support: Is the output supported by evidence that can be reconstructed and reviewed? Failure example: a finding is generated on input data that cannot be reproduced because originating records were deleted.
4. Authority to rely: Is the actor or system entitled to rely on the output for this purpose? Failure example: a delegated system takes an action the principal's mandate explicitly reserved for human decision.
5. Commitment boundary: Is the output about to produce a binding consequence? Failure example: an advisory output is automatically escalated to execution without a gateway requiring revalidation of admissibility at the commitment boundary.
6. Revalidation need: Have conditions changed such that admissibility must be re-resolved? Failure example: an output cleared at $T=0$ is relied upon at $T+6$ months after the regulatory environment changed; the system carries forward a prior clearance as if conditions were static.

This framework does not require treating machines as legal persons, conscious agents, or institutional interpreters. It treats machine outputs as artifacts whose admissibility depends on their relationship to a normative order at a specific moment in time.

8. Provenance Infrastructure and the Limits of Cryptographic Legitimacy

Blockchain-adjacent legal infrastructure raises the provenance-legitimacy distinction in its most acute form. Systems that combine cryptographic provenance with legal document management, norm application, or compliance workflows may generate a further confusion: that immutability, notarization, or verifiable history can solve the legitimacy problem. They cannot. They solve a different problem.

An immutable record can prove that an output existed at a given time, that a source was used, that a state transition occurred, that a chain of custody was preserved, or that a legal interpretation followed a documented path. This is valuable. It strengthens provenance and evidentiary reconstruction.

A blockchain record does not make a legal interpretation correct. It does not make an output admissible. It does not convert an inference into an institutional decision. It does not authorize reliance. Translating the tripartition from section 2.1: a blockchain record preserves normative propositions about the history of an output. It cannot convert those propositions into normative statements, because that conversion requires a normative act by a competent authority that no cryptographic protocol can perform.

The proper role of a provenance layer is therefore evidentiary and procedural:

- preserve source history;
- record interpretive lineage;
- make drift visible;
- support auditability;
- document state transitions;
- enable contestation;
- show when revalidation may be required.

Blockchain can preserve the history of legal reasoning, but it cannot by itself make that reasoning legally binding.

9. Implications for AI Governance

The simulation-realization distinction, reinforced by the legal logic analysis of section 2.1, has several implications for AI governance.

First, systems should not be evaluated only by output quality. They should be evaluated by the conditions under which outputs may be admitted into institutional reliance.

Second, auditability should not be conflated with admissibility. The EU AI Act's transparency requirements (Article 13) and logging mandates (Article 12) address the evidentiary dimension. They do not address the normative question of when a traced output may become institutionally binding. A complete governance architecture requires both layers.

Third, multi-step workflows require aggregate consequence review. Article 14 requires human oversight but frames it at the output level, not at the chain level. An agentic system may pass per-step oversight while the aggregate consequence of the chain was never reviewed as a whole.

Fourth, legal AI systems should distinguish four distinct output categories: advisory outputs (which inform but do not bind), evidentiary artifacts (which may be admitted in proceedings subject to standard evidential rules), institutional recommendations (which require a human principal to authorize before binding), and binding consequences (which require full runtime admissibility review). Annex III of the EU AI Act classifies high-risk AI systems by domain without requiring this internal taxonomy. That gap creates systematic misclassification risk.

Fifth, governance architectures should detect when revalidation is required. A system should not allow a prior clearance to carry a consequence through changed conditions without renewed admissibility review.

Sixth, the analytical tradition of legal logic provides the conceptual vocabulary that AI governance currently lacks: the distinction between normative propositions, normative statements, and norms; the difference between logical validity and systemic legal validity; and the temporal character of norm membership. Incorporating this vocabulary into AI governance frameworks would sharpen both the classification of AI outputs and the design of admissibility conditions.

10. Conclusion

AI systems can emulate legal reasoning, decision-making, and adjudicative form with increasing sophistication. But resemblance is not realization. A simulated decision is not yet an institutional decision. A plausible legal output is not yet admissible reliance. A traceable chain is not yet legitimate consequence.

The central problem for legal AI governance is therefore not whether machines can produce decision-like outputs. They can. The problem is whether and when those outputs may cross the boundary from technical event to institutional consequence.

That boundary requires a legal category beyond accuracy, explainability, auditability, and procedural validity. It requires institutional runtime admissibility: a continuing inquiry into whether the output can be recognized, reviewed, contested, relied upon, and allowed to bind within the relevant normative order. The category is not reducible to Hartian validity, Razian authority, evidential admissibility, or formal logical validity. It occupies a gap that each of those doctrines leaves open, a gap that the analytical tradition of legal logic allows us to specify with precision:

the gap between the logical correctness of a normative proposition and the institutional validity of a normative act.

The point is not that logic is irrelevant to law. Legal logic has always been, and remains, indispensable for the analysis of normative consistency, argumentative validity, and systemic coherence. The point is that legal logic operates within a normative system whose validity conditions are not reducible to computation. What the tradition reveals is not a blueprint for computational legal reasoning but a map of why the conditions for legal validity exceed the conditions for formal correctness.

Simulating the form of legal decision-making is not the same as performing the institutional act of deciding.

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